Low-power SAR ADCs: trends, examples and future

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***Abstract*— With the development of mobile devices and Internet-of-Things, the demand for low-power circuits has been growing rapidly. The Analog-to-Digital Converter (ADC) is a key building block in these systems. In this work, we review the progress of low-power ADCs over the years in terms of performance and limitations. From these limitations, it can be shown why sometimes very different approaches are used to minimize power consumption. Various state-of-the-art examples will be shown to illustrate the diversity of techniques and ideas within the field. Next, an outlook to the future is given by discussing various unsolved challenges and new opportunities.**

I. INTRODUCTION

Analog-to-Digital Converters (ADCs) are essential for many integrated circuits, as they are required to digitize physical (analog) signals to the digital domain. As such, they are used from high-speed applications like communication, to high-precision applications like sensing. Over the years, the efficiency of analog, digital, and mixed-signal circuits like ADCs has improved gradually. However, for analog functions (like amplifiers) this trend is relatively slow as their performance is limited by physical behavior, such as noise or matching. Digital circuitry has mostly improved thanks to technology scaling, following Moore's law. On the other hand, ADCs, as shown in Fig. 1 based on data from [1], have shown an efficiency improvement beyond 1000× over the past 2 decades, which is faster than technology scaling. The reason is threefold: first, most ADCs are not yet limited by physics; second, most ADCs benefit well from technology scaling; third, substantial circuit and architecture innovations took place that enabled the steep rate of improvement.

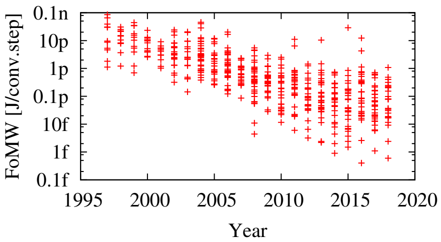

Fig. 1. ADC power efficiency (FoMW) over the years. Data from [1].

This work concentrates on the successive approximation register (SAR) ADC, as this has become the architecture of choice for medium-speed/medium-resolution requirements, where it managed to achieve the best efficiency amongst other ADC architectures, and it has a straightforward implementation. A basic SAR ADC topology is shown in Fig. 2. The analog input V_{in} is sampled on arrays of capacitors which also implement a charge-redistribution feedback DAC. The comparator, together with the logic and DAC perform a binary-search algorithm to approximate the sampled input in N steps, where N is the final resolution of the digital output code D_{out}.

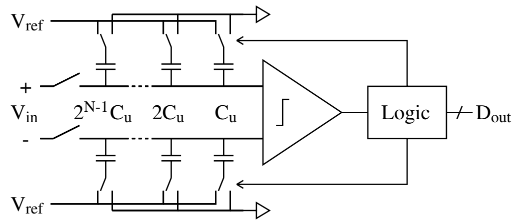

Fig. 2. Basic charge-redistribution SAR ADC.

The organization of this paper is as follows: Section II briefly describes various innovations that enabled efficiency improvement. An analysis of the power breakdown of SAR ADCs is also given. Next, Section III describes so-called hybrid ADCs, where the SAR principle is combined with other algorithms to further extend performance. Section IV discusses new challenges in the field of ADC design, as power efficiency is often not the bottleneck anymore. Several design examples are also given. Finally, conclusions are drawn in Section V.

II. POWER EFFICIENT SAR ADCS

A. Techniques for power efficient converters

Most innovations inside the SAR ADC focused at the switched-capacitor DAC and the comparator (see Fig. 2), as these are usually the most power consuming components. Thanks to the mixed-signal nature of an ADC, improvements can be made in the analog domain (circuit level), the digital domain (algorithm), or the physical level (layout). A few examples are given next.

The energy consumption of a switched-capacitor network is given in the form of $\alpha \cdot C \cdot V_{ref}^2$, where α depends on the network topology and way of switching, C is determined by the total capacitance of the network, and V_{ref} is the reference voltage. Note that this consumption has two components: the energy that is transferred from the source to the capacitors to enable the generation of particular voltages, and the energy that is lost during the charging process. While the first component is stored rather than dissipated, it is usually dissipated at a later phase when the capacitors are reset. To avoid unnecessary dissipation while charging, smart switching schemes are developed to lower the losses while maintaining the same functionality [2]. Similarly, to avoid losses due to discharge at reset, schemes are devised to re-use rather than reset this charge [3]. Both these methods aim to reduce α. Orthogonally to this, another approach is to minimize C down to the minimum required value, which often requires the use of non-standard layouts to achieve this, as in e.g. [4] or [5].

With respect to the comparator, circuit level improvements are possible to improve the efficiency (i.e., the ratio of energy used versus the noise level) [6]. However, this is ultimately limited by physics, and thus does not allow unlimited savings. Nonetheless, algorithmic techniques can further improve efficiency as not all comparator decisions in a binary search are equally critical, thus energy can be saved for non-critical cases, for instance by means of noise-tolerant design [7] or using selective majority-voting [8].

B. Power breakdown analysis

The above examples show how energy can be saved in the DAC and comparator, but it does not clarify which component is actually dominant in the overall design. This section discusses the power breakdown of an exemplary SAR ADC under different design assumptions. For this analysis, the measured data of a 10bit SAR ADC with a conventional switching scheme using 0.25fF unit capacitors in 65nm CMOS [9] is taken as a starting point. The power consumption is divided in four parts: DAC, comparator, SAR logic, and basic. Basic stands here for the constant consumption required by the overhead circuits that do not scale with resolution. From this known data, the power consumption for other target resolutions is extrapolated under three assumptions as shown in Fig. 3:

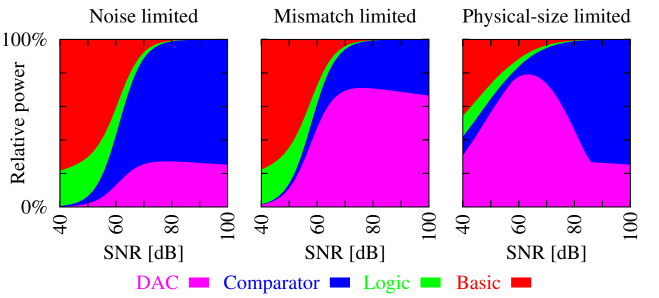

Fig. 3. Estimated SAR ADC power breakdown, based on [9], with capacitors limited by noise, mismatch, or physical size.

1) Assuming there is only a noise constraint for DAC and comparator. In this case, the total DAC capacitance and thus its power scale with 4^N and the comparator power scales with $N{\cdot}4^N$, in order to meet the SNR requirement. As shown, here the comparator power dominates, even though the DAC is using a basic conventional switching scheme.

2) Assuming the DAC unit capacitor is sized for matching constraints, i.e. to guarantee sufficient DAC linearity. In this case, the DAC capacitors typically become larger, and now the DAC power dominates over the comparator. Here the focus should be on saving DAC power, or calibration can be applied to size down the DAC towards the noise limited situation.

3) Assuming the DAC unit capacitor is physically limited to a minimum value (in this example 2fF), due to the physical implementation. This has no impact for high SNR, as large capacitors are needed here anyhow to satisfy noise, thus this situation is the same as scenario 1). However, for low SNR, due the minimum value of 2fF, the DAC capacitance becomes much larger than required for matching or noise. As a result, the DAC dominates overall power for medium SNRs, and the focus should be on saving DAC power by e.g. switching schemes or by implementing smaller capacitors.

As an overall conclusion, for >10bit resolution, the DAC and comparator dominate power. The exact proportion between them strongly depends on the assumption of how the capacitors are sized. Ultimately, the fundamental limit is noise rather than mismatch or physical value. In such cases, the comparator is usually most critical. For <10bit resolution, even the basic and logic power can play a substantial role.

III. Hybrid SAR converters

This section illustrates several hybrid ADC architectures, in which the efficient SAR principle is combined with pipelining (for speed and resolution) or noise-shaping (for resolution).

A. Pipelined SAR ADCs

Conventionally, pipelined converters use a pipeline of low-resolution flash ADCs with interstage amplifiers to amplify their residue, as shown in Fig. 4a, resulting in high-speed and high-resolution converters. However, a lot of power is spend on the many accurate interstage amplifiers. With the progress of technology, SAR ADCs became efficient and fast, leading to the pipelined SAR architecture [10], as sketched in Fig. 4b. While still being fast, the efficient SAR ADC and the reduction of the amount of amplifiers greatly improved power efficiency. As a next step, research focused on implementing interstage gain stages with better efficiency, e.g. zero-crossing based, ring, or dynamic amplifiers [11]. Thanks to these new amplifiers, the conventional architecture now also becomes competitive again, especially for very high speed applications.

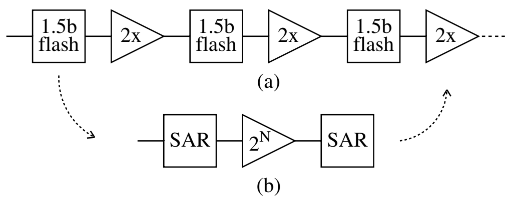

Fig. 4. Pipelined ADCs with many flash stages (a) or few SAR stages (b).

B. Noise-shaping SAR ADCs

High-resolution ADCs are often made with $\Sigma\Delta$ modulators (Fig 5a), as their loop filter and feedback create noise-shaping to enable high resolution. Moreover, the low-resolution internal quantizer and feedback DAC can achieve a very high linearity, e.g. by using DEM techniques. The SAR principle can be applied in two ways: either it can be used as the quantizer (ADC) in a $\Sigma\Delta$ modulator (Fig 5a), or a regular SAR can be made first, after which noise-shaping is only applied to the residue of the SAR ADC (Fig 5b) [12], [13]. The latter system is called a noise-shaping (NS) SAR ADC. Compared to the $\Sigma\Delta$ modulator, the noise shaping is only applied to the residue rather than the entire system, while there is no explicit feedback DAC required as the SAR's own DAC can be used inherently. While this allows a more simple implementation, the implication is also that DAC kT/C noise is not shaped, thus large capacitors are required to achieve sufficiently low noise. Also, many NS SAR ADCs use a (semi-)passive loop filter, which results in an increase of capacitance or an increase of noise. Lastly, DAC mismatch cannot easily be solved with DEM techniques anymore, as the DAC's resolution is too high. As a result, calibration is required to achieve sufficient linearity, or mismatch error shaping is required [14].

Currently, most noise-shaping SAR ADCs use modest loop-filter orders and modest oversampling factors. Apart from the resolution gain thanks to oversampling, the extra gain thanks to noise-shaping is typically limited to about 1 or 2 bit of resolution. For even higher resolutions, the conventional $\Sigma\Delta$ modulator is still a better choice as it can achieve high linearity more easily, and it can employ more aggressive noise-shaping.

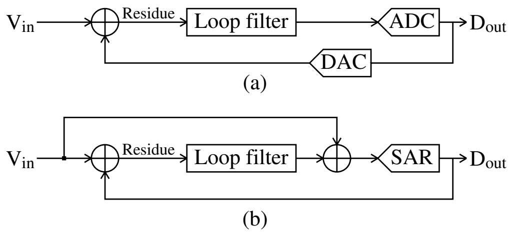

Fig. 5. $\Sigma\Delta$ modulator (a) and Noise-Shaping SAR ADC (b).

IV. SAR ADC CHALLENGES AND EXAMPLES

Due to the rapid efficiency improvements over the years, the ADC is often not anymore the bottleneck at system level. Therefore, research should shift to new challenges, a few of which are described here together with design examples.

A. Linearity

Matching of the binary-scaled array of capacitors in the DAC (Fig. 2) is critical for the overall linearity. Due to matching limitations, achieving a very high resolution SAR ADC is cumbersome. While calibration techniques [15] or mismatch error shaping techniques [14] have been successfully implemented, such techniques are not yet mainstream part of SAR ADC design, due to their complexity or limitations. Simple yet effective solutions are still demanded, also to avoid overdesigning the DAC capacitors for matching only, which causes a severe penalty in chip area and efficiency (Fig. 3).

B. Chip area

The chip area of a SAR ADC is relatively large, mostly due to the value and large number of unit capacitors required for the DAC. Efficient layout implementations can help solving this problem. In [5], rather than using 2^N unit capacitors to compose an N bit DAC, the N binary-scaled elements are made as sketched in Fig. 6, where 3 binary scaled capacitors $C_{0,1,2}$ are made by scaling the length of their metal strips. However, due to end effects this does not give accurate matching. To solve this, compensation capacitors $C'_{0,1,2}$ are switched oppositely to the main capacitors, such that the effective value of each capacitor is purely defined by the length difference of the main and compensation element. In this way, end effects are compensated, accurate matching is achieved, and the number of layout elements is reduced from 2^N to $2{\cdot}N$.

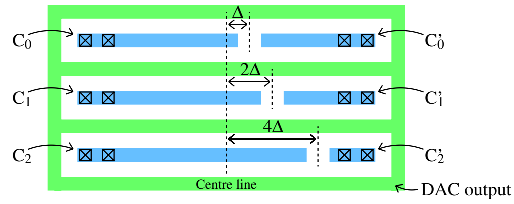

Fig. 6. Unit-length capacitor scaling, according to [5].

C. Time-domain processing

With the scaling of technology, voltage headroom is going down while frequencies go up. This led to interest to represent signals in the time domain rather than the voltage domain (Fig. 7), where the magnitude is for instance represented by the length of a pulse. While this can be advantageous, it does not necessarily push the boundaries in performance. First of all, the voltage headroom is shrinking, but noting that from 0.18μm CMOS down to 14nm FinFET, the supply changed from 1.8V to 0.8V, only 7dB of dynamic range is lost in the voltage domain. Further, assuming a timing jitter of 0.1ps (state-of-the-art for ADCs), the time-domain representation is not only limited by the aperture error when sampling, but the sample rate also limits the maximum time, and thus the dynamic range, of the time-domain signal (Fig. 8). As a result, the time-domain representation cannot cover the same design space as the voltage-domain representation, but may offer benefits in terms of chip area and technology portability.

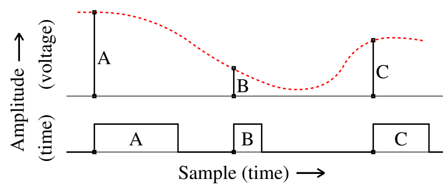

Fig. 7. Voltage-domain and time-domain representations.

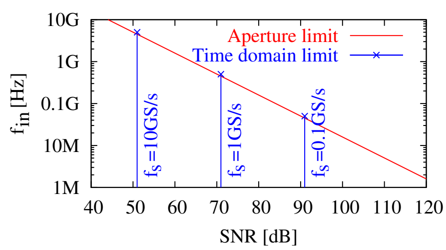

Fig. 8. Aperture limit and time-domain processing limit (jitter = 0.1ps).

D. System integration

Even though the ADC core became highly efficient over the years, this is not the case for the surrounding circuits. In practice, the input signal driver and reference driver that need to charge the DAC capacitors to V_{in} and V_{ref}, respectively, often consume more than the entire ADC. Since these drivers are usually conventional analog blocks, limited by noise requirements or drive-strength requirements, they tend not to scale well, and thus hamper overall system efficiency. Co-design of the ADC with its drivers and applying mixed-signal techniques to these otherwise analog circuits can help to overcome this challenge. Two examples are described here.

Fig. 9 shows a temperature sensor, where a bridge network composed of temperature-sensitive resistors produces a temperature-dependent voltage which is directly driving the ADC [16]. The bridge is duty-cycled, such that the consumed energy for a measurement approximates the minimum energy $C{\cdot}V^2$ that is required to charge the DAC. By removing the active signal driver in front of the ADC and only using a passive network, the overall system becomes more simple and more efficient. Also, thanks to discrete-time operation, its power consumption now scales proportional to the measurement rate.

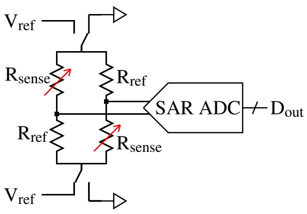

Fig. 9. All-dynamic SAR-based temperature sensor, based on [16].

As a second example, [17] discusses the co-integration of the reference driver with a SAR ADC. Normally, this driver is either large in chip area or high in power consumption, in order to stabilize the DAC voltage quickly between each cycle of the SAR algorithm. Instead, [17] uses a small and low-power driver, while its resulting inaccuracy is compensated by implementing a non-linear correction function in the DAC. In this way, the combined system achieves an overall better efficiency as compared to isolated designs.

E. Early and adaptable digitization

Software Defined Radio, where the ADC is placed as close as possible to the antenna, has been in development for a long time. While this approach is not as efficient as classical radio architectures, the progress of ADC performance and efficiency makes it gradually more attractive to pursue this direction. Also, the inclusion of other functionalities within the ADC helps to achieve this target.

Applications at lower speeds, such as sensor interfaces, can also benefit from the flexibility of ADCs and their ability to be moved closer to the frontend, as it enables higher flexibility and adaptivity of the entire system. For instance, in [18], a flexible and self-adaptive sense-and-compress system is build around the sensor from [16].

V. Conclusion

In this work, various limitations affecting SAR ADC efficiency were discussed, and example techniques to enhance performance were illustrated. The analysis shows that the critical component depends strongly on the assumptions made in the design, like limitations due to noise, mismatch or physical constraints. Hybrid SAR converters were introduced, and their performance area was indicated, showing that they can achieve higher speed or resolution compared to standard SAR ADCs, but typically below the absolute performance of conventional pipelined or $\Sigma\Delta$ converters. Lastly, several recent challenges and example solutions were described, revealing that integration with other blocks, more adaptability, and earlier digitization become feasible and create new degrees of freedom and opportunities in the design.

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